

1. Smart Agriculture & Plant Care Assistant

Integrates sensor monitoring with online information to achieve precision farming.

- Application Scenario: Students connect an AMB82-mini to a soil moisture sensor and a water pump.
- Technical Implementation:

Use `/analogread` to read soil moisture levels.

Use `/search` to query water requirements for specific plants (e.g., succulents) under current weather conditions.

AI combines search results and sensor data to proactively ask:

“Soil moisture is low and today is hot. Turn on the water pump for 10 seconds?” If confirmed, execute `/on`.

Use `/vision` to periodically capture leaf images to detect pests or plant diseases.

2. Industrial Inspection & Equipment Maintenance

Uses vision reasoning and connectivity to reduce manual inspection costs.

- Application Scenario: Monitor gauges or safety indicators in laboratories or training factories.
- Technical Implementation:

Schedule `/vision` tasks to capture pressure gauges or meters.

AI analyzes images; if the needle enters the red zone, immediately use `/search` for emergency troubleshooting manuals.

Send alerts via Telegram and suggest executing `/off` to shut down main power pins (Pin 24/25).

Use `soul.md` to define a “Senior Safety Engineer” persona for professional tone guidance.

3. Smart Retail & Product Presenter

Enhances customer experience through multimodal recognition and interaction.

- Application Scenario: Installed near product displays to identify items and provide real-time information.
- Technical Implementation:

Customers show products to the camera, triggering `/vision` for recognition.

AI uses `/search` to retrieve reviews, promotions, or price comparisons.

Use `/pwm` to control display lighting (Pin 13), increasing brightness when customers approach.

Use `memory.md` to record frequently asked questions and analyze product trends.

4. Home Care & Smart Emergency Alert System

Enhances environmental awareness and remote communication for safety.

- Application Scenario: Assistive device for elderly people living alone.
- Technical Implementation:
 - Use `/digitalread` for an emergency button (Pin 12); when pressed, send Telegram alerts immediately.
 - Family members can use `/still` anytime to check home status.
 - AI uses `/vision` to recognize medicine bottles and remind correct medication usage.
 - Use `/on` to control medication box indicators or emergency alarms.

5. Environmental Monitoring & Carbon Reduction Assistant

Real-time data analysis with automated control.

- Application Scenario: Air quality monitoring in classrooms or offices.
- Technical Implementation:
 - Use `/analogread` to monitor PM2.5 or CO₂ sensors.
 - If thresholds are exceeded, use `/search` to check outdoor air quality and allergy forecasts.
 - If outdoor air is good, suggest opening windows; otherwise, activate air purifier via `/on`.
 - Use `/memory` to track historical power consumption vs air quality trends.

6. Smart Classroom Interactive Assistant

Combines AI dialogue, vision, and device control to enhance teaching efficiency.

- Application Scenario: AI assistant in classrooms supporting teachers.
- Technical Implementation:
 - Use `/vision` to recognize blackboard or projection content and generate summaries.
 - Use `/search` for teaching resources such as formulas or historical context.
 - AI may suggest: “Turn on display light (`/pwm` pin 13) to improve projection clarity?”
 - Use `/digitalread` to detect classroom buttons (e.g., help request button).
 - Use `memory.md` to record student misconceptions and build class learning models.

7. Smart Parking Management System

Improves parking efficiency using sensing and vision.

- Application Scenario: Campus or community parking lots.
- Technical Implementation:
 - Use `/vision` to detect occupied or empty parking spaces.
 - Use `/search` for real-time traffic and parking availability.
 - AI reports: “Slot 3 is available, turn on guide light (/on pin 25)?”
 - Use `/pwm` to adjust indicator brightness (empty / occupied / reserved).
 - Send real-time status via Telegram.

8. AI Lab Safety & Chemical Monitoring System

Monitors laboratory environments for safety risks.

- Application Scenario: Chemistry labs or technical classrooms.
- Technical Implementation:
 - Use `/analogread` to monitor gas sensors (e.g., VOC, alcohol).
 - If abnormal, use `/search` for hazard information and safety procedures.
 - Trigger Telegram alerts and recommend activating ventilation via `/on`.
 - Use `/vision` to detect leaks or unsafe behavior (e.g., no gloves).
 - Use `soul.md` as a “Lab Safety Officer” persona for authoritative tone.

9. Smart Home Energy Management System

Optimizes household energy usage through AI decisions.

- Application Scenario: Home or dormitory energy monitoring system.
- Technical Implementation:
 - Use `/analogread` to measure current consumption.
 - AI uses `/search` to check peak/off-peak electricity pricing.
 - Suggest turning off high-power devices via `/off` during peak hours.
 - Use `/pwm` to reduce lighting brightness for energy savings.
 - Use `/memory` to analyze long-term energy usage patterns.

10. Disaster Early Warning & Emergency Response System

Combines sensing, vision, and external data for rapid response.

- Application Scenario: Earthquake, fire, or typhoon emergency systems.
- Technical Implementation:
 - Use `/analogread` to monitor smoke, temperature, or vibration sensors.
 - Use `/search` for local disaster alerts and response guidelines.
 - Use `/vision` to detect fire, smoke, or structural collapse.
 - Automatically trigger `/on` to activate alarms and escape indicators.
 - Broadcast alerts via Telegram.

11. Animal & Livestock Care System

Improves automation and health monitoring in farming environments.

- Application Scenario: Fish farms, chicken coops, or pet care systems.
- Technical Implementation:
 - Use `/vision` to monitor animal activity and abnormal behavior.
 - Use `/search` for environmental requirements (temperature, feed).
 - Trigger alerts if abnormal inactivity is detected.
 - Use `/pwm` to control feeding machines or lighting systems.
 - Use `memory.md` to maintain individual health records.

12. AI Edge Surveillance Agent

Autonomous intelligent monitoring node.

- Application Scenario: Warehouses, campuses, or offices.
- Technical Implementation:
 - Use `/vision` to detect people, motion, or intrusions.
 - Use `/search` to verify whether activity is normal (e.g., cleaning schedule).
 - Trigger `/still` for evidence capture and upload to Telegram during anomalies.
 - Use `/on` to activate warning lights or alarms.
 - Use `memory.md` to improve future anomaly detection.

13. Smart Aquarium Management System

Maintains optimal aquatic environments via automation.

- Application Scenario: Home aquariums or educational ecology tanks.
- Technical Implementation:
 - Use `/analogread` to monitor pH and water temperature.
 - Use `/search` for fish species water requirements.
 - Adjust aeration via `/pwm` when water quality is abnormal.
 - Use `/vision` to detect fish count and abnormal behavior.
 - Use `/on` for feeding systems and UV sterilization lights.
 - Use `memory.md` for long-term water quality tracking.

14. Smart Exhibition Guide Assistant

Provides personalized real-time exhibition guidance.

- Application Scenario: Museums, tech exhibitions, or campus showcases.
- Technical Implementation:
 - Use `/vision` to identify exhibits visitors are viewing.
 - Use `/search` to provide historical background or research updates.
 - Use `/pwm` to adjust spotlight brightness based on visitor attention.
 - Use `soul.md` as a professional museum guide persona.
 - Use `memory.md` to track popular exhibits for curators.

15. Smart Gym Equipment Monitor

Ensures safety and provides training feedback.

- Application Scenario: Community gyms or school fitness rooms.
- Technical Implementation:
 - Use `/vision` to evaluate posture correctness and injury risk.
 - Use `/search` for correct exercise techniques.
 - Use `/analogread` to detect abnormal vibration.
 - Send real-time coaching feedback via Telegram.
 - Use `/on` to activate equipment status lights after sanitation.

16. Smart Library Assistant

Improves library automation and reader experience.

- Application Scenario: School or public libraries.
- Technical Implementation:
 - Use `/vision` to identify books by cover or spine.
 - Use `/search` for reviews and recommendations.
 - Use `/analogread` for humidity and temperature monitoring.
 - Use `/pwm` for adaptive reading light control.
 - Use `memory.md` to track popular books for acquisition planning.

17. Unmanned Store Assistant System

Enhances automation and customer safety in unattended stores.

- Application Scenario: Campus stores or vending shops.
- Technical Implementation:
 - Use `/vision` to track product removal.
 - Use `/search` for stock and replenishment suggestions.
 - Use `/digitalread` for weight sensors to detect item removal.
 - Trigger `/still` for evidence capture in abnormal cases.
 - Use `memory.md` for sales heatmap analysis.

18. Smart Greenhouse Automation System

Maximizes crop yield via precise environmental control.

- Application Scenario: Agricultural education or urban farms.
- Technical Implementation:
 - Use `/analogread` for temperature, humidity, light, and CO₂ levels.
 - Use `/search` for optimal crop growth conditions.
 - Suggest activating shading systems or misting via `/on` or `/pwm`.
 - Use `/vision` for crop health monitoring (disease, yellowing).
 - Use `memory.md` for weekly growth logs.

19. AI Companion Robot Assistant

Provides emotional support and daily assistance.

- Application Scenario: Therapy centers, autism training, elderly care.
- Technical Implementation:
 - Use `soul.md` for a warm and patient personality.
 - Use `/vision` for facial emotion recognition.
 - Adjust interaction style based on emotional state.
 - Use `/pwm` for mood-based lighting interaction.
 - Use `memory.md` for personalized user preferences.
 - Send alerts via Telegram if user is unresponsive for long periods.

20. Smart Warehouse Inventory System

Automates inventory tracking and reduces human error.

- Application Scenario: School storage rooms or logistics centers.
- Technical Implementation:
 - Use `/vision` for item recognition and counting.
 - Use `/search` for stock comparison and shortage detection.
 - Use `/digitalread` for entry/exit timestamp tracking.
 - Trigger Telegram alerts with `/still` for unauthorized movement.
 - Use `memory.md` to predict replenishment cycles.

21. Smart Traffic Intersection Assistant

Improves pedestrian and vehicle safety.

- Application Scenario: Campus roads or community intersections.
- Technical Implementation:
 - Use `/vision` to detect pedestrians crossing.
 - Use `/on` to control warning lights for vehicles.
 - Use `/analogread` for vibration-based traffic estimation.
 - Use `/search` to predict peak traffic patterns.
 - Use `memory.md` for traffic risk analysis reports.

22. Smart Restroom Management System

Improves hygiene and facility efficiency.

- Application Scenario: Public restrooms in schools or buildings.
- Technical Implementation:
 - Use `/digitalread` for stall occupancy detection.
 - Send real-time status via Telegram.
 - Use `/analogread` for ammonia gas monitoring.
 - Use `/pwm` to adjust lighting based on usage.
 - Use `memory.md` for peak usage analysis.

23. Food Safety Inspection Assistant

Supports food safety monitoring in kitchens and restaurants.

- Application Scenario: School kitchens or small restaurants.
- Technical Implementation:
 - Use `/vision` to detect spoilage or contamination.
 - Use `/search` for food safety alerts or blacklist data.
 - Use `/analogread` for refrigerator temperature monitoring.
 - Use `/on` to activate backup cooling systems.
 - Use `soul.md` for a strict food inspector persona.

24. Smart After-School Tutoring System

Provides AI-powered learning assistance and assessment.

- Application Scenario: After-school programs or home learning.
- Technical Implementation:
 - Use `/vision` to analyze handwritten homework.
 - Use `/search` for solutions and explanations.
 - Generate targeted practice questions based on mistakes.
 - Use `memory.md` for student progress tracking.
 - Use `soul.md` for a patient and encouraging tutor persona.